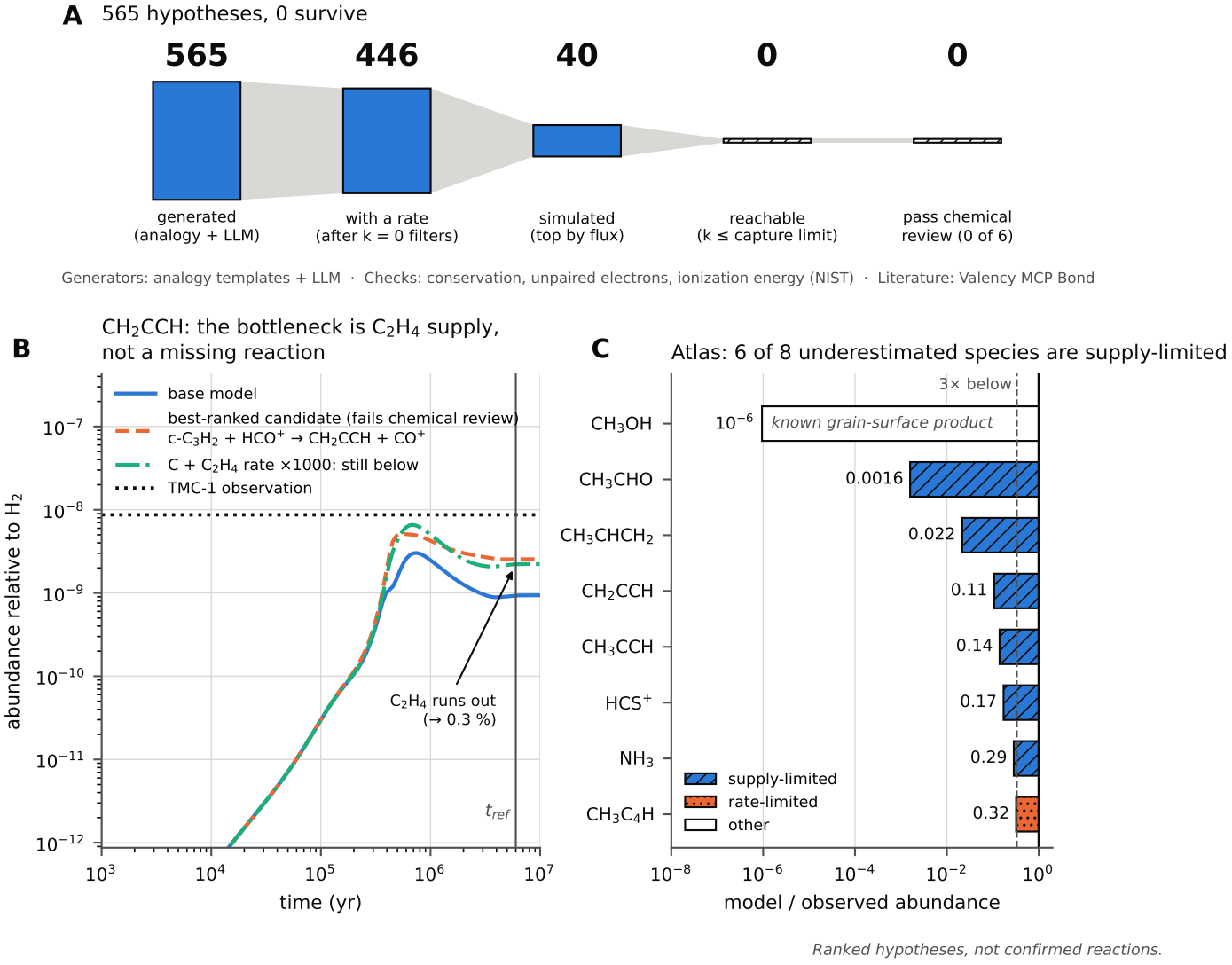

A
565 hypotheses, 0 survive
565
446
40
0
0
generated
(analogy + LLM)
with a rate
(after k = 0 filters)
simulated
(top by flux)
reachable
(k ≤ capture limit)
pass chemical
review (0 of 6)
Generators: analogy templates + LLM · Checks: conservation, unpaired electrons, ionization energy (NIST) · Literature: Valency MCP Bond
B
CH2CCH: the bottleneck is C2H4 supply,
not a missing reaction
base model
best-ranked candidate (fails chemical review)
c-C3H2 + HCO+ → CH2CCH + CO+
C + C2H4 rate ×1000: still below
TMC-1 observation
C2H4 runs out
(→ 0.3 %)
tref
abundance relative to H2
time (yr)
C
Atlas: 6 of 8 underestimated species are supply-limited
3× below
CH3OH
10−6
known grain-surface product
CH3CHO
0.0016
CH3CHCH2
0.022
CH2CCH
0.11
CH3CCH
0.14
HCS+
0.17
NH3
0.29
CH3C4H
0.32
supply-limited
rate-limited
other
model / observed abundance
Ranked hypotheses, not confirmed reactions.